

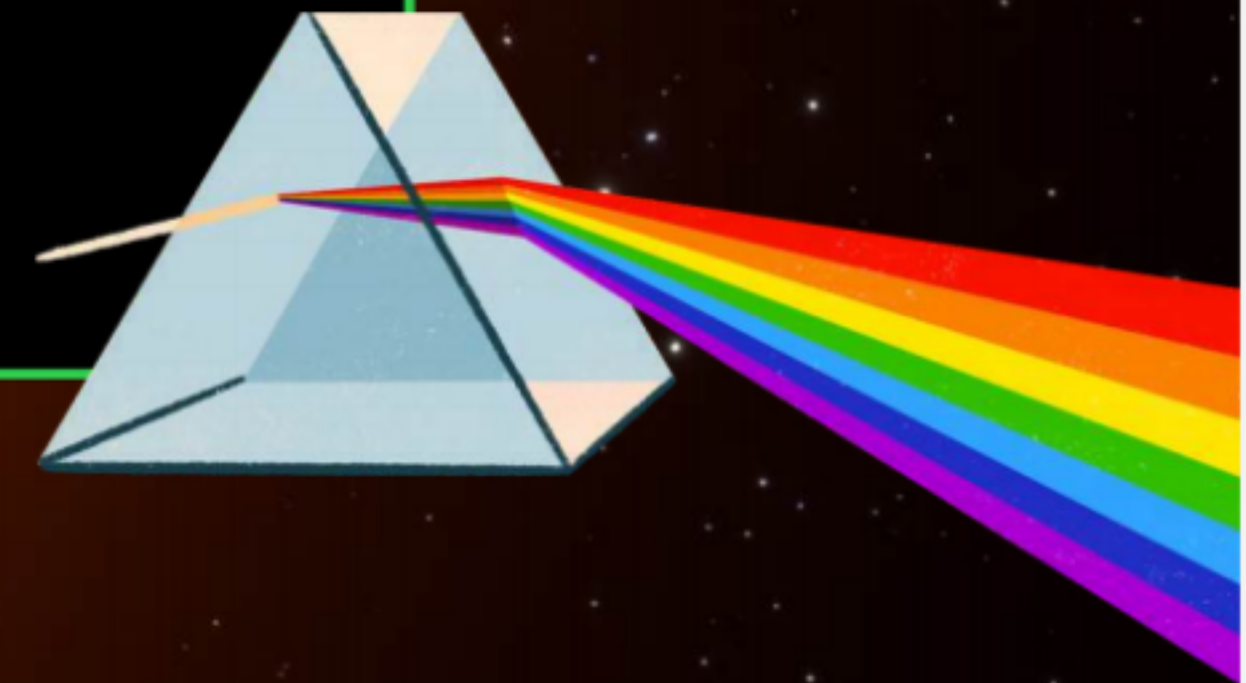
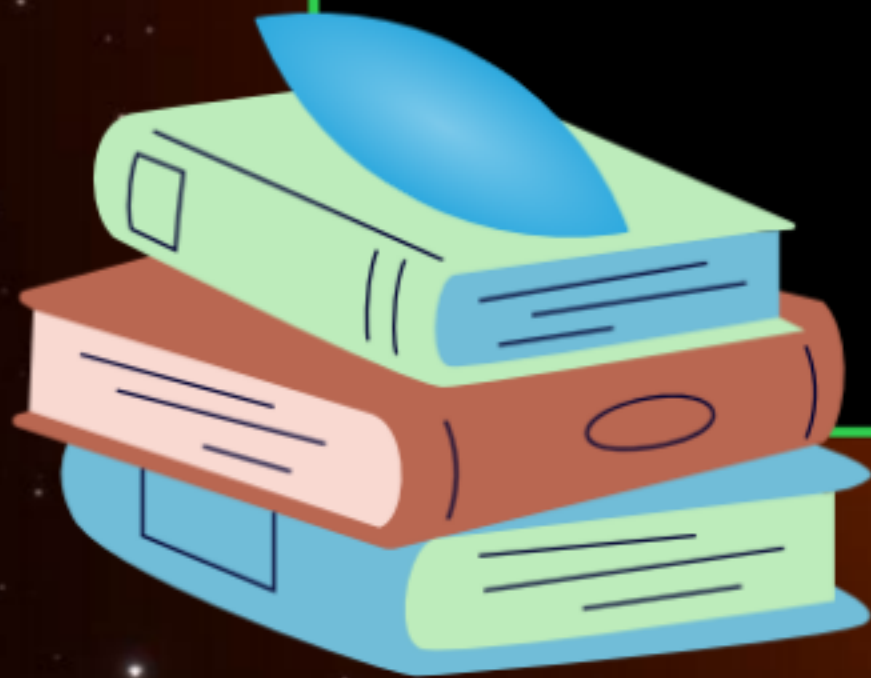
L5

आरंभ
BATCH

CLASS X - SCIENCE

LIGHT

PRASHANT KIRAD



Refraction through Glass Slab

When a ray of light passes through a glass slab (a transparent rectangular block), it bends twice:

1. At the first surface (air to glass):

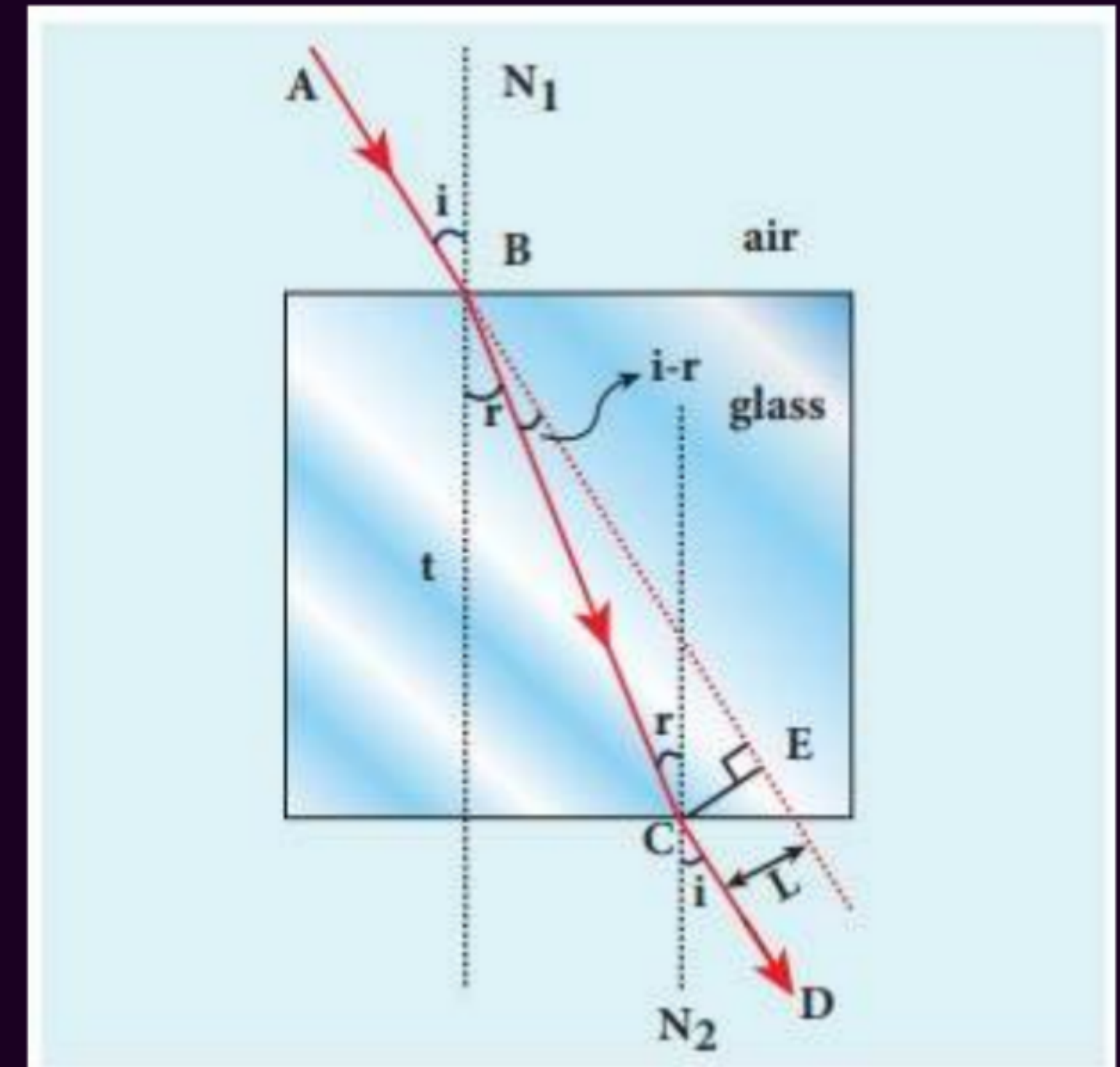
- Light slows down and bends towards the normal.

2. Inside the slab:

- Light travels straight but at a slower speed.

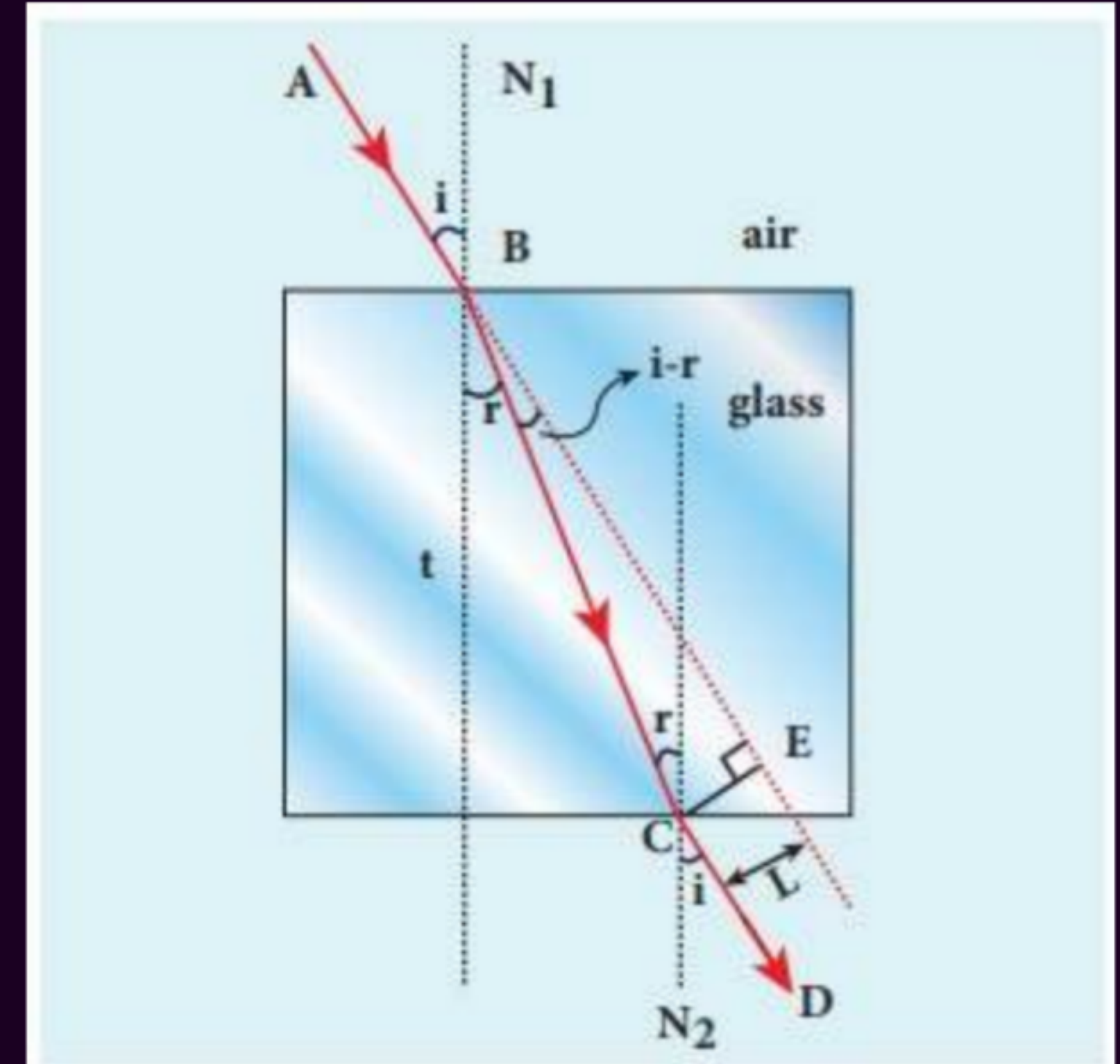
3. At the second surface (glass to air):

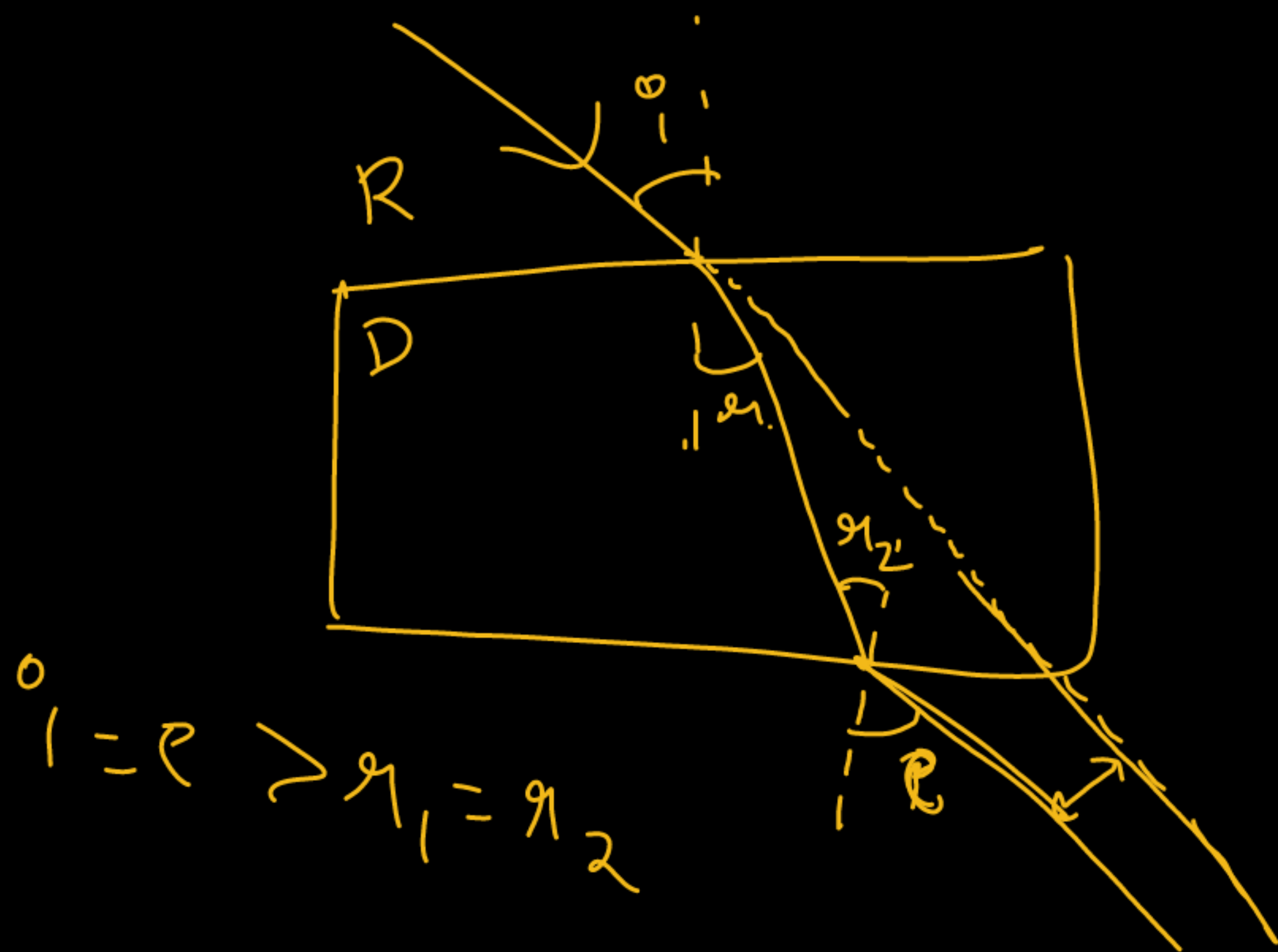
- Light speeds up and bends away from the normal.



The incident ray and the emergent ray (the one coming out) are parallel to each other but shifted sideways.

[This sideways shift is called ***lateral displacement***.]





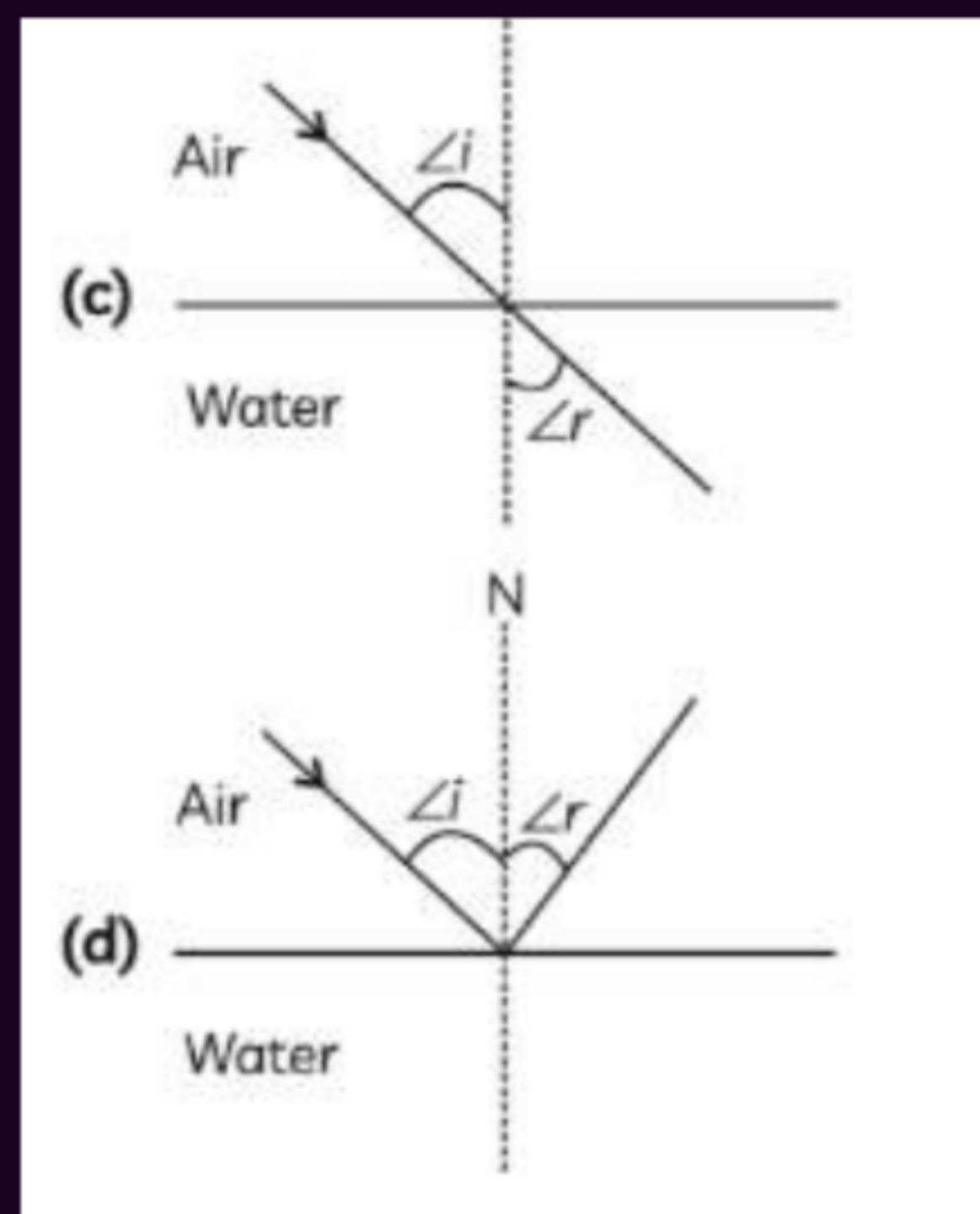
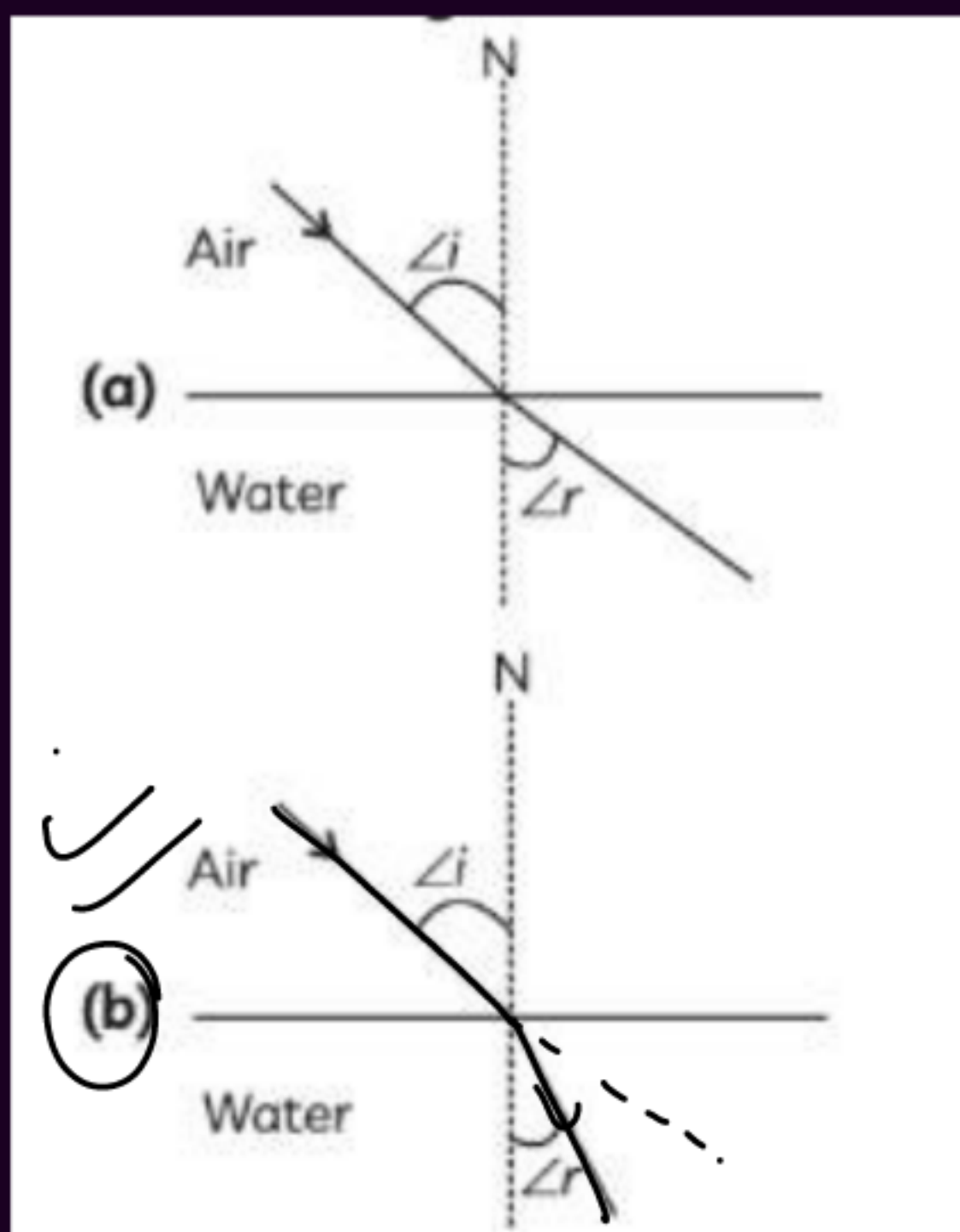
$$\frac{\sin i}{\sin r} = n$$

$n \propto \text{density}$
 $\hookrightarrow \propto \text{Bending}$

$$n \propto \frac{1}{v}$$

Soch ke batao

Q. When a ray of light passes from air into water at an angle of incidence of 45° , which diagram correctly represents the path of the refracted ray?



Soch ke batao

- Q. The apparent depth of a pond is less than its real depth due to:
- Jo humhe di kh Rha
- (a) Reflection of light
 - (b) Dispersion of light
 - ☒ (c) Refraction of light
 - (d) Scattering of light



Soch ke batao

$$V = 1 \times 6$$

Q. The refractive index of glass for violet light is greater than that for red light because:

- (a) violet light travels slower in glass than red light
- (b) red light bends more than violet light
- (c) wavelength of violet light increases in glass
- (d) frequency of red light changes more rapidly

$$\mu_v > \mu_{Red}$$



Soch ke batao

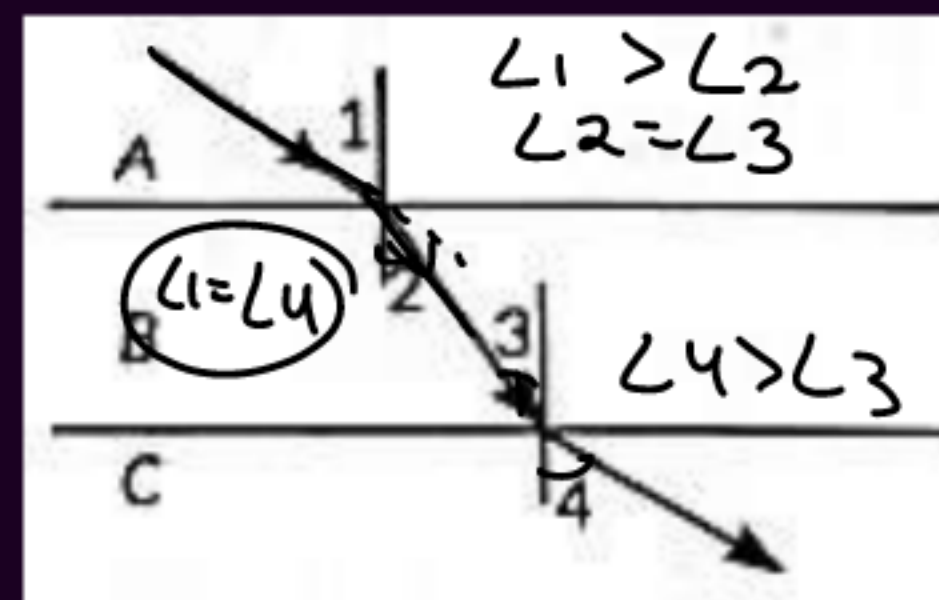
Q. A ray of light is incident as shown. If A, B and C are three different transparent media, then which among the following options is true for the given diagram?

(a) $\angle 1 > \angle 4$

(b) $\angle 1 < \angle 2$

(c) $\angle 3 = \angle 2$

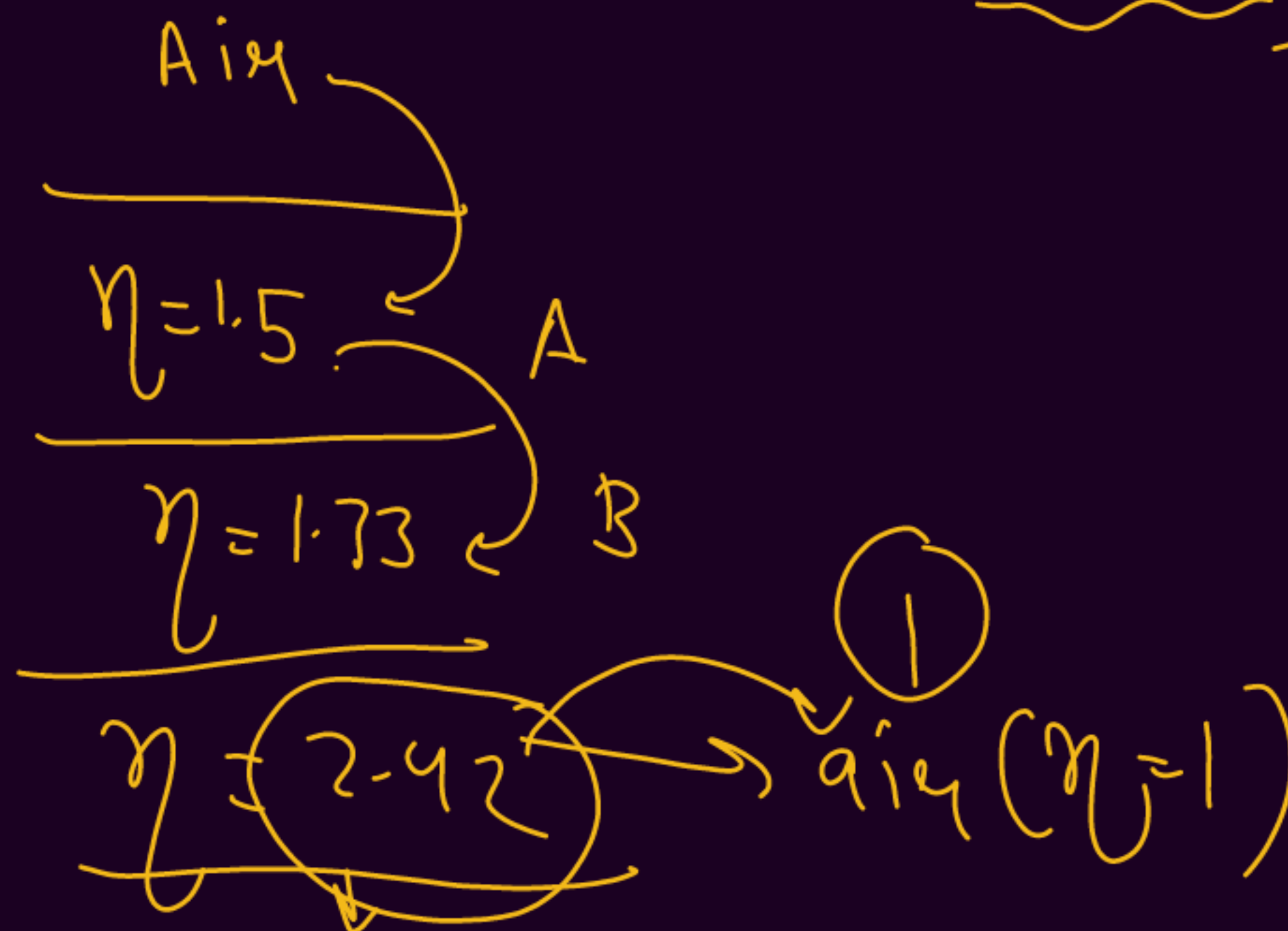
(d) $\angle 3 > \angle 4$



Soch ke batao

Q: A ray of light starting from air passes through a medium A of refractive index 1.50, enters medium B of refractive index 1.33 and finally enters medium C of refractive index 2.42. If this ray emerges out in air from C, then for which of the following pairs of media is the bending of light the least? (CBSE 2021, 2022)

- (a) Air – A
- (b) A – B
- (c) B – C
- (d) C – Air



Soch ke batao

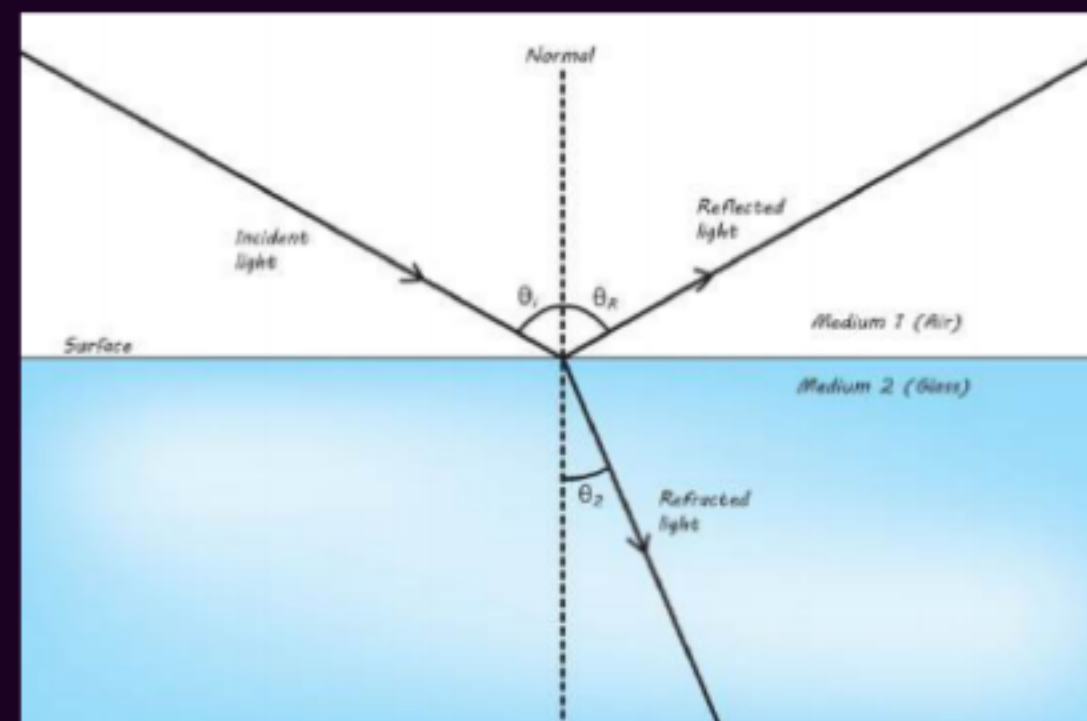
- Q. (a) Why does a pencil appear bent when placed in water? ✓
(b) Draw a ray diagram showing refraction through a rectangular glass slab.

✓ (a) A pencil appears bent when placed in water due to refraction of light.

Light rays coming from the pencil bend at the boundary between water and air because light travels at different speeds in different media. This change in direction makes the pencil appear bent at the water surface.

(b) Refraction through a rectangular glass slab

- When a ray of light passes through a rectangular glass slab, it bends towards the normal on entering the slab and away from the normal on leaving it. The emergent ray is parallel to the incident ray, but laterally displaced.



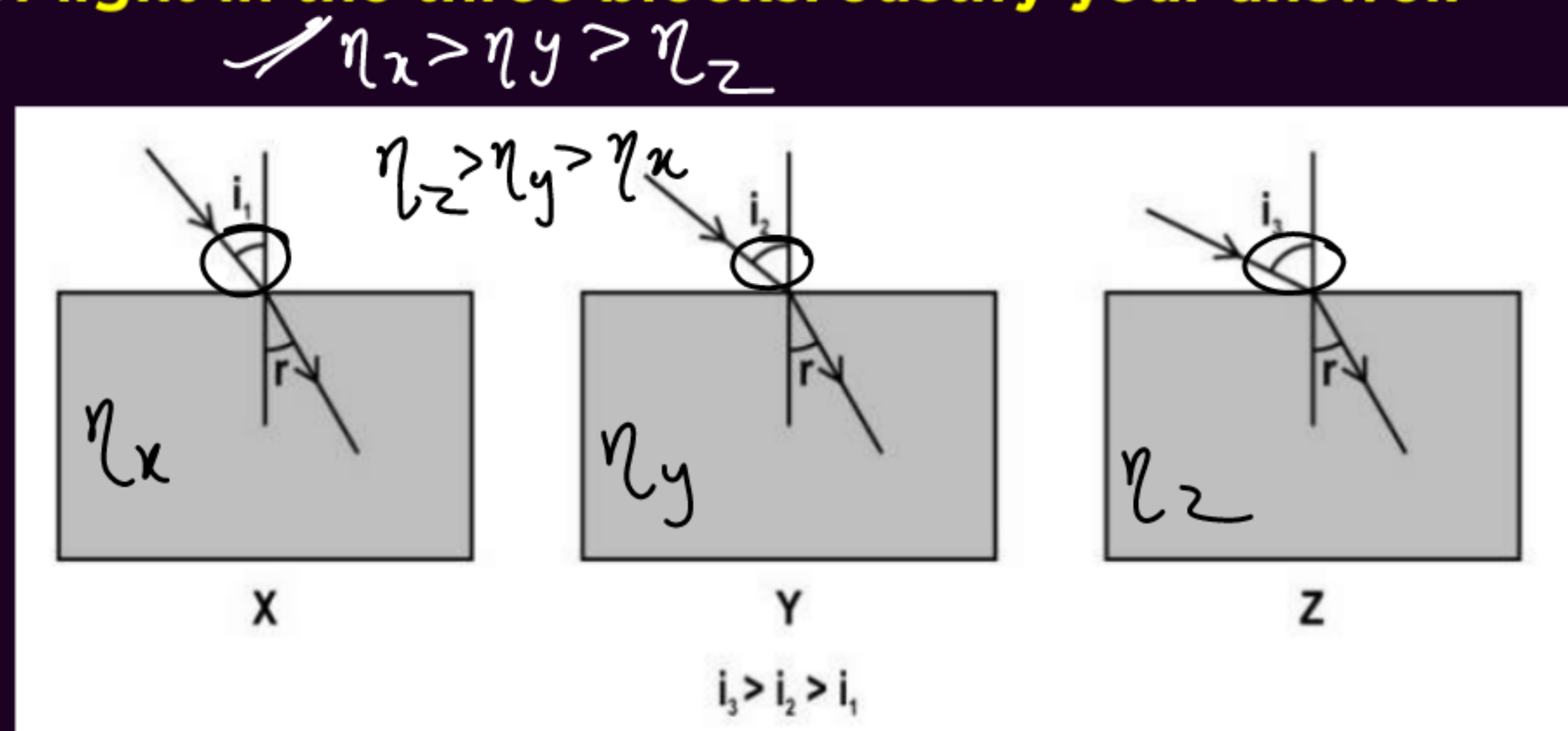
Soch ke batao

Q. The refractive index of a medium 'x' with respect to a medium 'y' is $\frac{3}{2}$, and the refractive index of medium 'y' with respect to medium 'z' is $\frac{4}{3}$. Find the refractive index of medium 'z' with respect to medium 'x'. If the speed of light in medium 'x' is $3 \times 10^8 \text{ m s}^{-1}$, calculate the speed of light in medium 'y'. (2020)



Soch ke batao

Q. The image below shows the refraction of light in three transparent rectangular blocks X, Y and Z, made of different materials, when they are placed in air. The angle of incidence is different in each case, but the angle of refraction is the same in all three blocks. Compare the speed of light in the three blocks. Justify your answer.



Q Why does the coin placed at the bottom of a bucket filled with water appear raised and displaced from its actual position?

Answer: This happens due to refraction of light. When light travels from water to air, it bends, so the coin appears raised and shifted from its real position.

Activity 9.7

- Place a coin at the bottom of a bucket filled with water.
- With your eye to a side above water, try to pick up the coin in one go. Did you succeed in picking up the coin?
- Repeat the Activity. Why did you not succeed in doing it in one go?
- Ask your friends to do this. Compare your experience with theirs.



Due to refraction of light, the coin appears raised and displaced from its actual position.

Absolute Refractive Index

If medium 1 is vacuum or air, then the refractive index of medium 2 is considered with respect to vacuum. This is called the absolute refractive index of the medium.

$$n = \frac{c}{v}$$

It is denoted by n and is given by the formula:

$$n_m = \frac{\text{Speed of light in air}}{\text{Speed of light in the medium}} = \frac{c}{v}$$

where:

- c = speed of light in vacuum ($\approx 3 \times 10^8$ m/s)
- v = speed of light in the medium

Relative Refractive Index

The refractive index of medium 2 with respect to medium 1 is called the relative refractive index.

$$n_{21} = \frac{n_2}{n_1}$$

It is denoted by n_{21} and is given by the formula:

$$n_{21} = \frac{\text{Speed of light in medium 1}}{\text{Speed of light in medium 2}} = \frac{v_1}{v_2}$$

where:

- v_1 = speed of light in medium 1
- v_2 = speed of light in medium 2

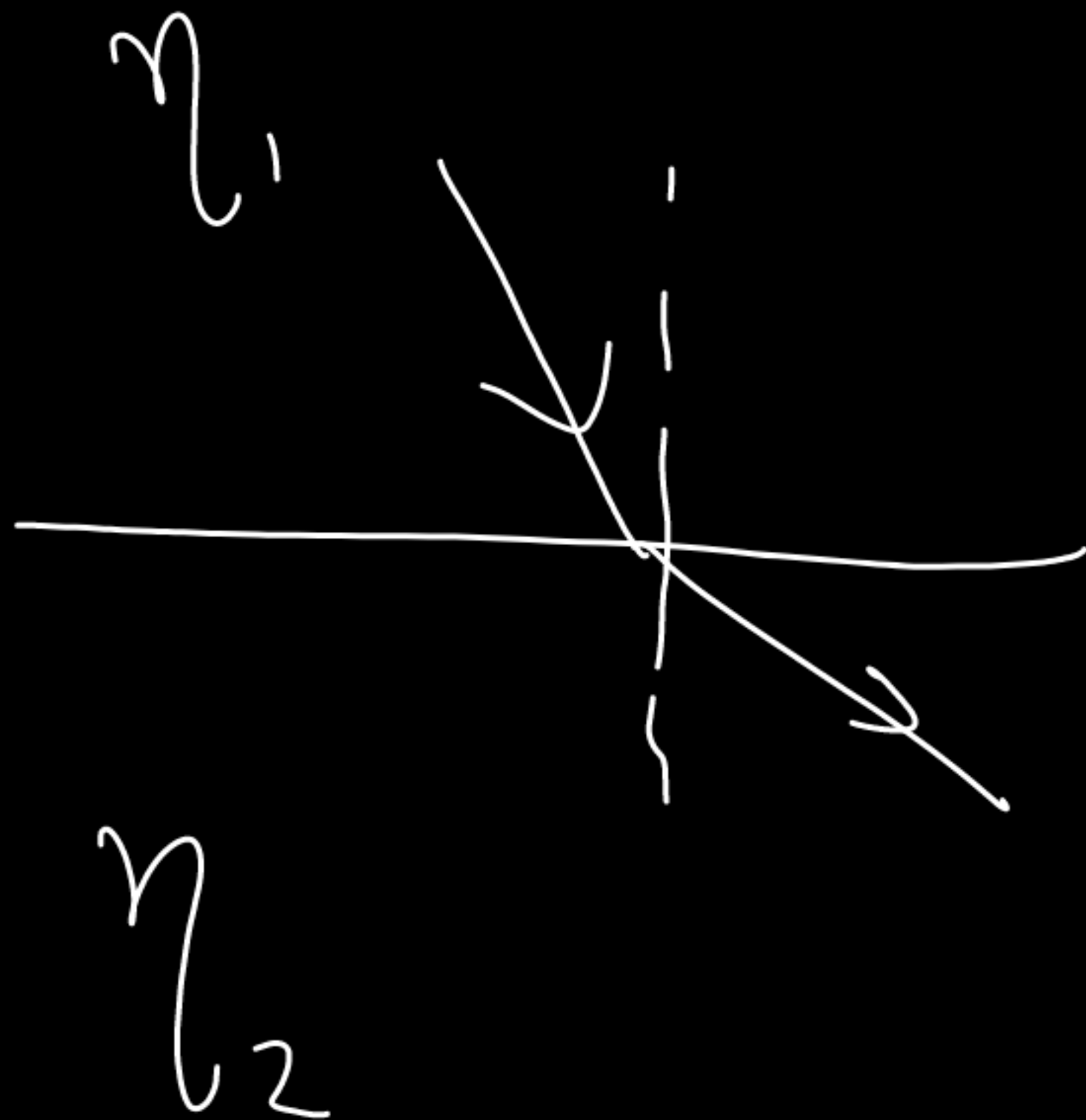
④

④

$$\eta_{21} = \frac{\eta_2}{\eta_1} = \frac{V_1}{V_2}$$

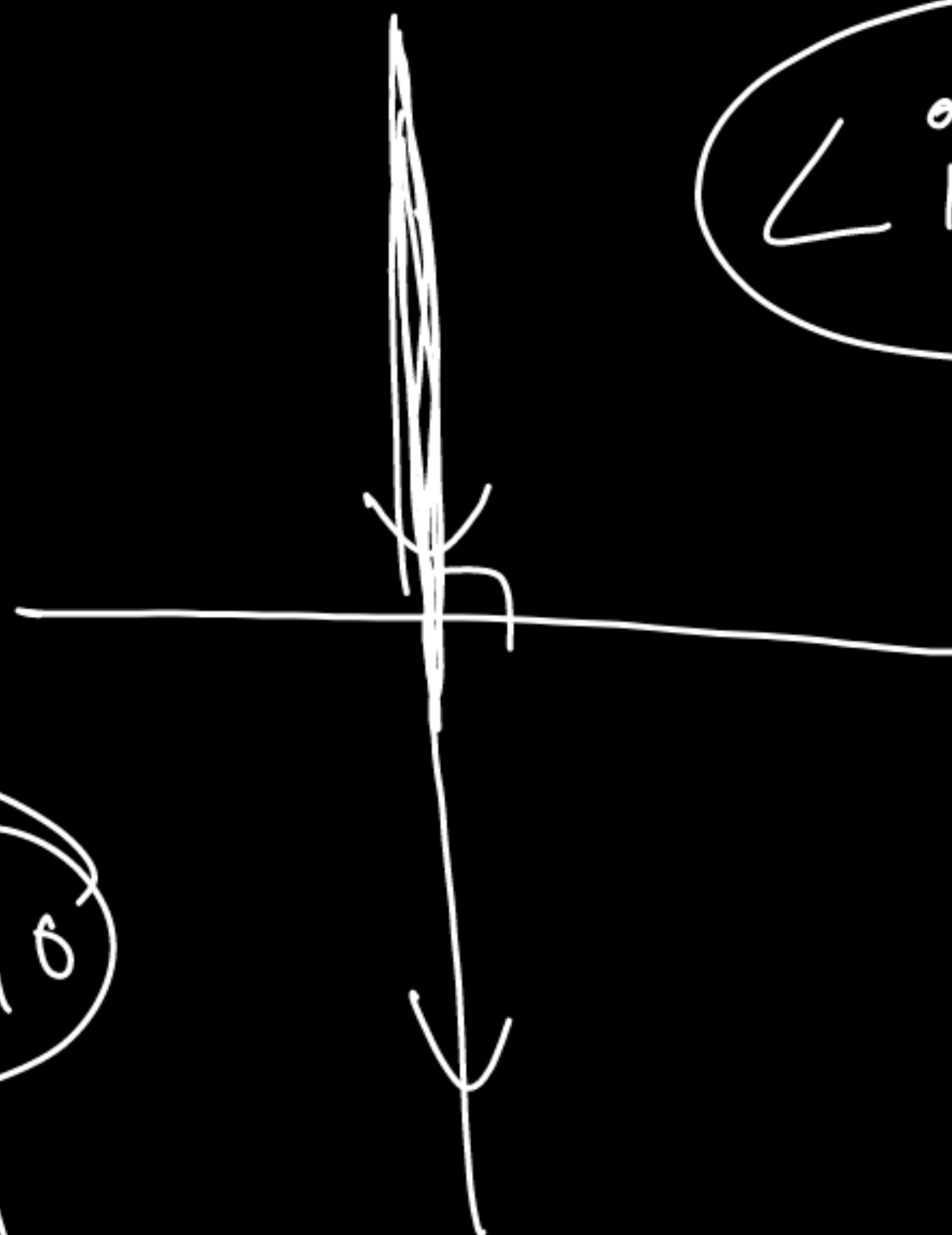
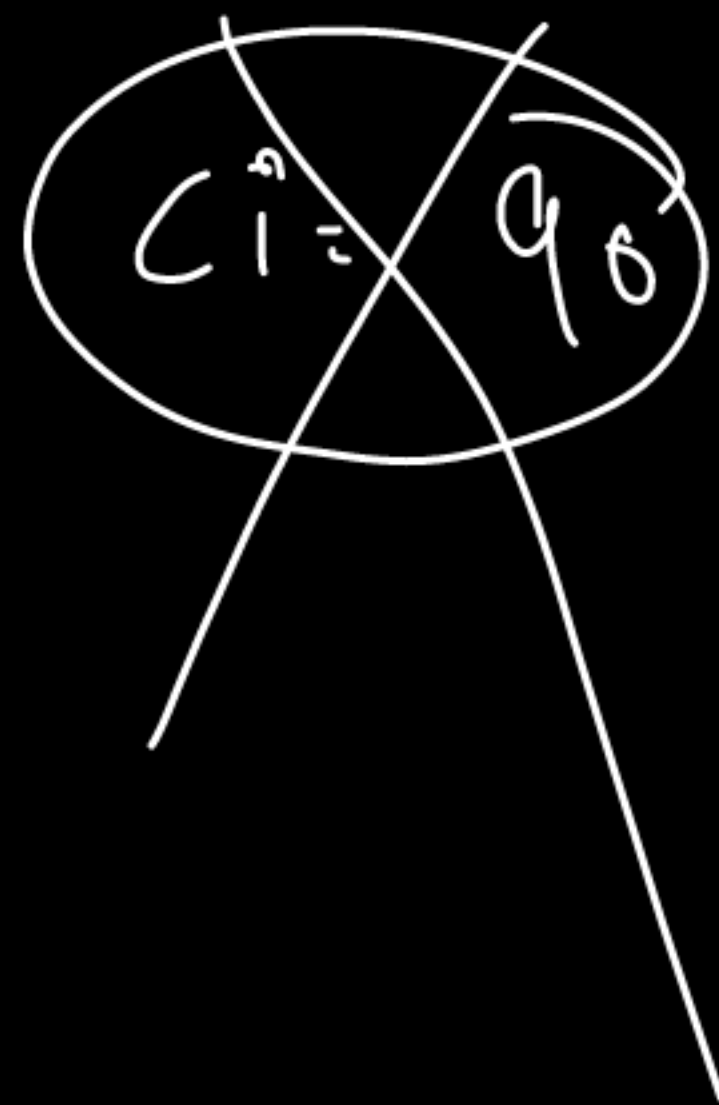
$$\left(\eta \propto \frac{1}{V} \right)$$

$$\eta_2 \text{ with respect to } \eta_1 = \frac{\eta_2}{\eta_1}$$



$$\eta_R = \frac{\eta_2}{\eta_1}$$

$$\eta_{21} = \frac{\eta_2}{\eta_1}$$



$$\angle i = ? \quad 0^\circ$$



Soch ke batao

$$v = 1$$

Q. The refractive index of a medium is 1.5. The speed of light in that medium will be:

- (a) Greater than speed in vacuum
- (b) Equal to speed in vacuum
- (c) Less than speed in vacuum
- (d) Zero



Soch ke batao

Q. Absolute refractive index of water and glass is $\frac{4}{3}$ and $\frac{3}{2}$, respectively. If the speed of light in glass is 2×10^8 m/s, the speed of light in water is: [CBSE 2024, 2025]

- (a) 2.25×10^8 m/s
- (b) 4×10^8 m/s
- (c) 3×10^8 m/s
- (d) 2.5×10^8 m/s

$$\frac{n_2}{n_1} = \frac{v_1}{v_2}$$

$$\frac{n_w}{n_g} = \frac{v_g}{v_w}$$



Soch ke batao

Q. If the absolute refractive ^{Index} indices of two media X and Y are $\frac{6}{5}$ and $\frac{4}{3}$ respectively, then the refractive index of Y with respect to X will be: [CBSE 2025]

- (a) $\frac{10}{9}$
- (b) $\frac{9}{10}$
- (c) $\frac{9}{8}$
- (d) $\frac{8}{9}$

$$\eta = \frac{\frac{4}{3} \times \frac{5}{6}}{\frac{6}{5}} = \frac{20}{18} = \frac{10}{9}$$

$$\frac{\eta_y}{\eta_x} = \frac{\frac{4}{3}}{\frac{6}{5}}$$



Soch ke batao

Q. The refractive index of medium A is 1.5 and that of medium B is 1.33. If the speed of light in air is 3×10^8 m/s, what is the speed of light in medium A and B respectively? (2022)

- (a) 2×10^8 m/s and 1.33×10^8 m/s
- (b) 1.33×10^8 m/s and 2×10^8 m/s
- (c) 2.25×10^8 m/s and 2×10^8 m/s
- (d) 2×10^8 m/s and 2.25×10^8 m/s

$$\mu_A = \frac{c}{v}$$

$$1.5 = \frac{3 \times 10^8}{v}$$

$$v_A = 2 \times 10^8 \text{ m/s}$$

$$\mu_B = \frac{c}{v}$$

$$1.33 = \frac{3 \times 10^8}{v}$$

$$v = \frac{3}{1.33} \times 10^8$$



Soch ke batao

HW

Q. Light enters from air to glass having a refractive index of 1.50. What is the speed of light in the glass? The speed of light in vacuum is $3 \times 10^8 \text{ m s}^{-1}$.



Soch ke batao

Q. The refractive index of diamond is 2.42 while that of glass is 1.5.

(a) Which substance slows down light more? → diamond

(b) In which substance will refraction be greater?

(c) Why does diamond sparkle more brilliantly than glass?

→ Bends more light



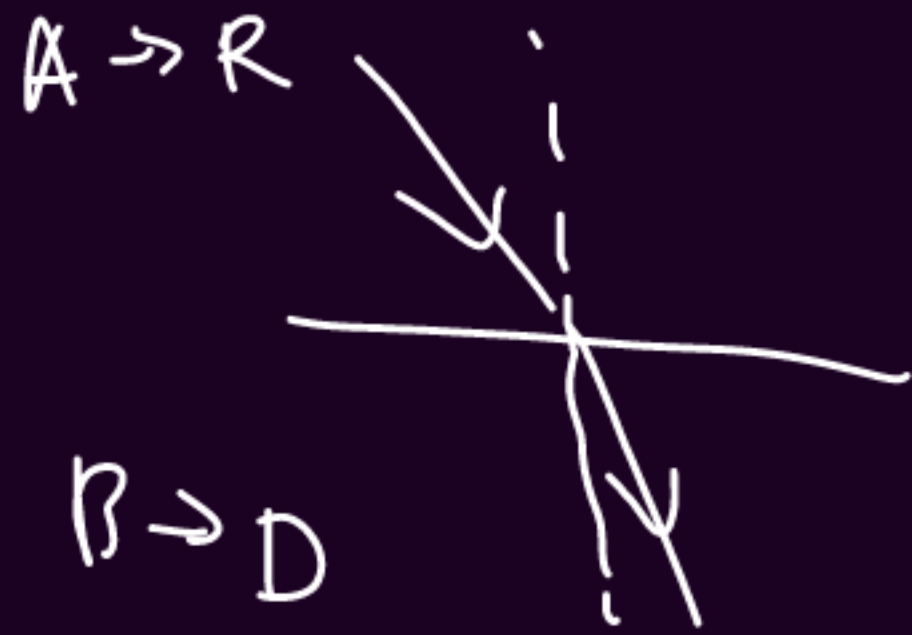
Soch ke batao

Q. The speed of light in medium A is 2×10^8 m/s and in medium B is 1.25×10^8 m/s.

(a) Calculate refractive indices of both media.

(b) Which medium is optically rarer? $\rightarrow \eta$ less

(c) How will light bend when travelling from A to B?



$$\eta_A = \frac{c}{v} = \frac{3 \times 10^8}{2 \times 10^8} = 1.5 \checkmark$$

$$\eta_B = \frac{3 \times 10^8}{1.25} = 2.4$$



Soch ke batao

Q. A ray of light passes from air into a transparent medium. The angle of incidence is 60° and the angle of refraction is 35° .

- (a) Calculate the refractive index of the medium.
- (b) Determine the speed of light in the medium.
- (c) State whether the medium is optically denser or rarer than glass ($n = 1.5$).



Soch ke batao

R-D

Q. The refractive indices of three media are given below:

Medium	Refractive Index
A	1.6
B	1.8
C	1.5

Handwritten notes on the table:

- From A to B: Towards Normal
- From B to C: Away

Diagram showing light rays:

- A ray from A to B bends towards the normal.
- A ray from B to C bends away from the normal.

A ray of light is travelling from A to B and another ray is travelling from B to C.

- (a) In which of the two cases does the refracted ray bend towards the normal?
- (b) In which case does the speed of light increase in the second medium? Give reasons for your answer.

Soch ke batao

$$\frac{\mu_A}{\mu_B} = \frac{1.33}{1.44}$$

Q. During an experiment in the laboratory, Tushar derived refractive indices of four media, i.e., A, B, C and D as shown in the table below:

Media	Max A	B	C	Min D
Refractive Index	1.33	1.44	1.52	1.65

Help Tushar to identify the media in which the speed of light is:

- (i) minimum and
- (ii) maximum?

Give reasons for your answer.

Also, find the refractive index of medium A with respect to medium B.

Soch ke batao

Q. A light ray travels from ^Rwater ($n = 1.33$) into ^Dglass ($n = 1.52$). If the angle of incidence in water is 40° , calculate the angle of refraction in glass.

Also determine:

- (a) Relative refractive index of glass with respect to water
- (b) Change in speed of light as it enters glass

$$\frac{\sin i}{\sin r} = \frac{1.52}{1.33}$$

$$\frac{\sin 40}{\sin r} = \frac{1.52}{1.33}$$

$$\eta = \frac{1.52}{1.33}$$

$$\frac{\sin i}{\sin r} = \eta$$

$$\frac{1.33 \sin 40}{1.52} = 1.52 \sin r$$



Soch ke batao

HW

Q. A student performs an experiment with a rectangular glass slab of refractive index 1.5. A ray of light is incident at 45° . The angle of refraction inside the slab is 28° .

- (a) Verify Snell's law for the given data.
- (b) Calculate the speed of light in glass.
- (c) Explain why the emergent ray becomes parallel to the incident ray even after refraction occurs twice.



Soch ke batao



- Q. (a) The absolute refractive indices of water and glass are $\frac{4}{3}$ and $\frac{3}{2}$ respectively. If the speed of light in glass is 2×10^8 m/s, find the speed of light in water.
- (b) The refractive index of glass is 1.5. Calculate the speed of light in glass.
- (c) If the refractive index of medium A is greater than B, in which medium will light travel faster? Why?



Soch ke batao

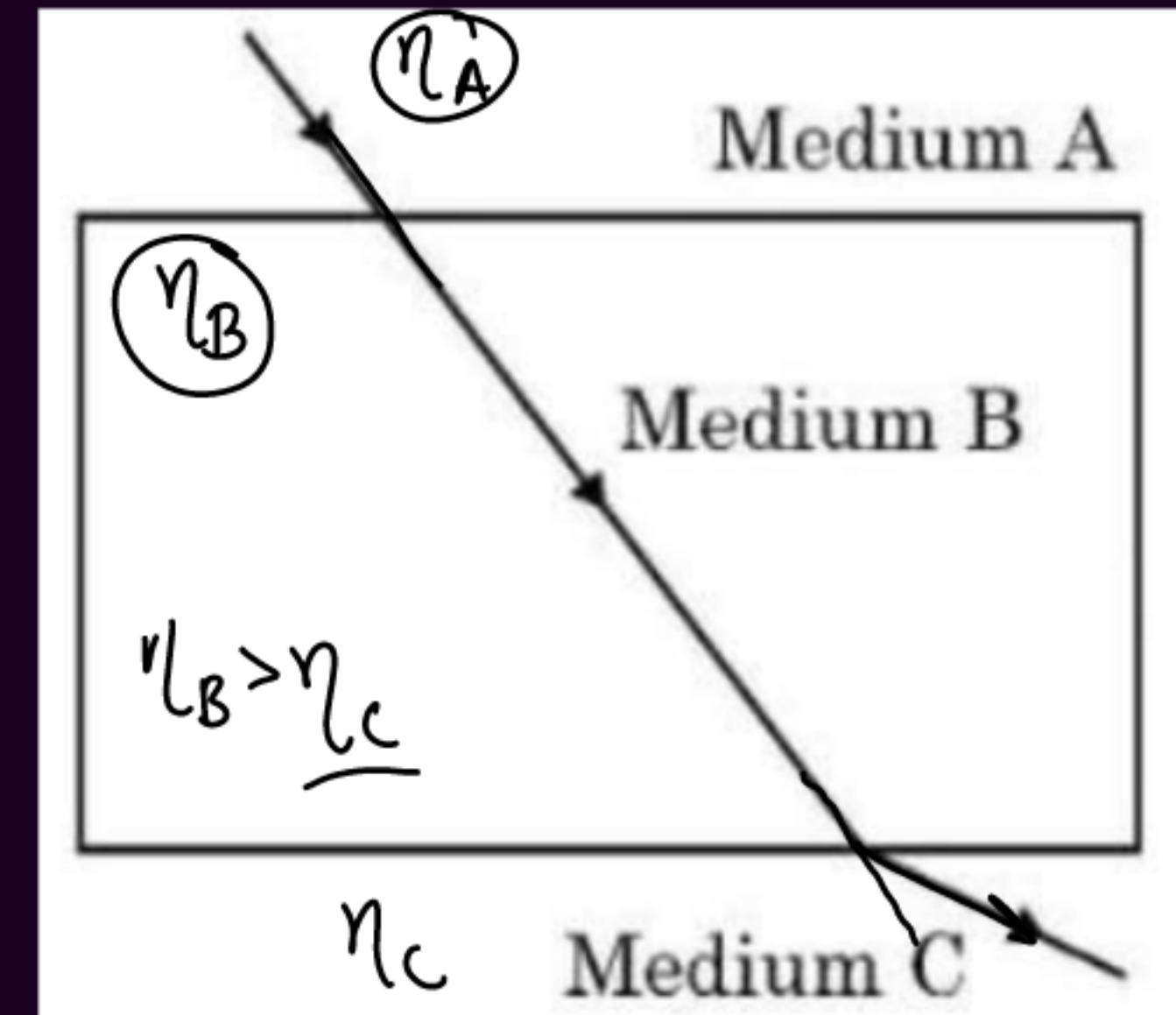
Q. (a) Observe the following diagram and compare (i) speed of light and (ii) optical density of the three media A, B, and C. Also give justification for your answer of any one of the two cases in terms of refractive indices of A, B, and C.

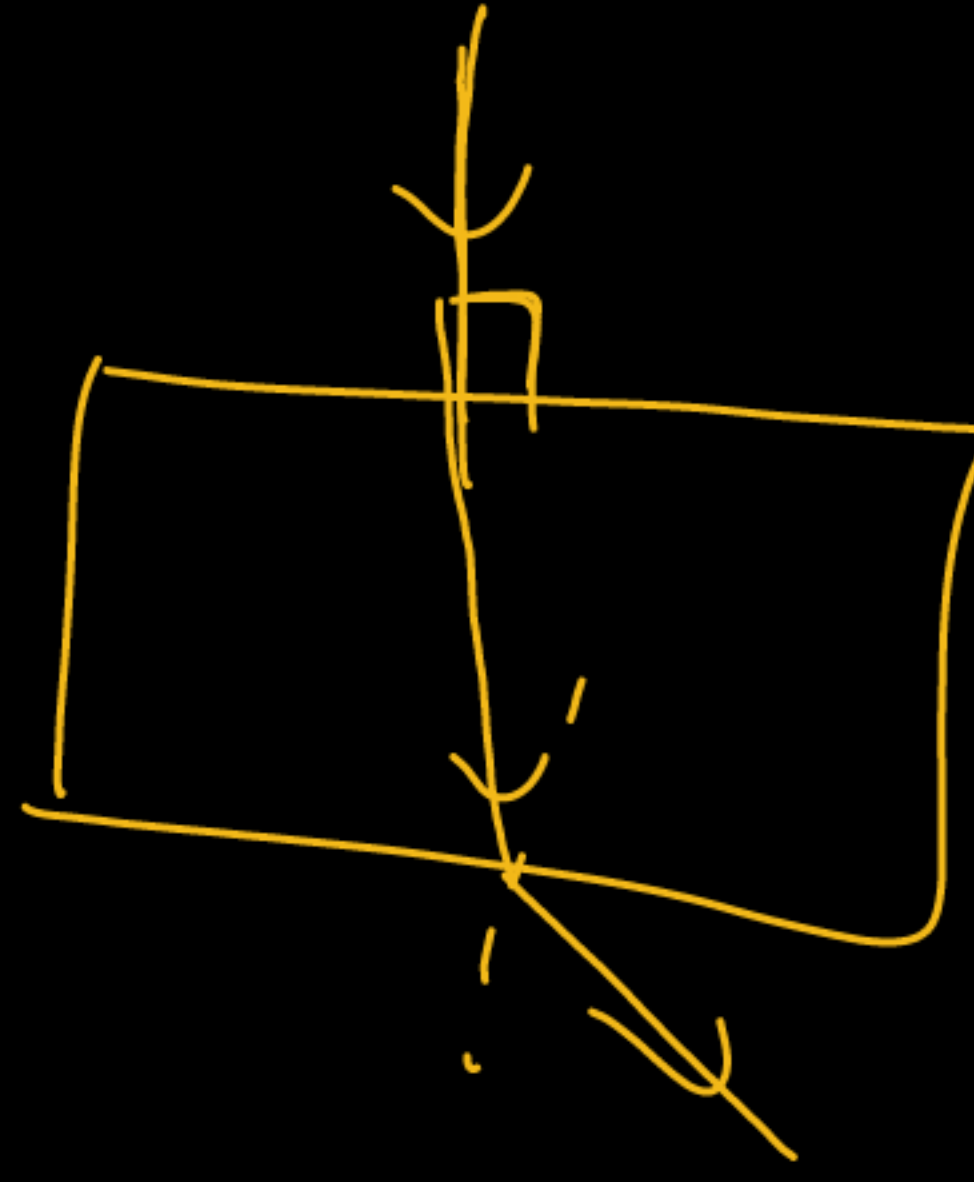
(b) Redraw the path of a ray of light through the three media, if the ray of light starting from medium A falls on medium B:

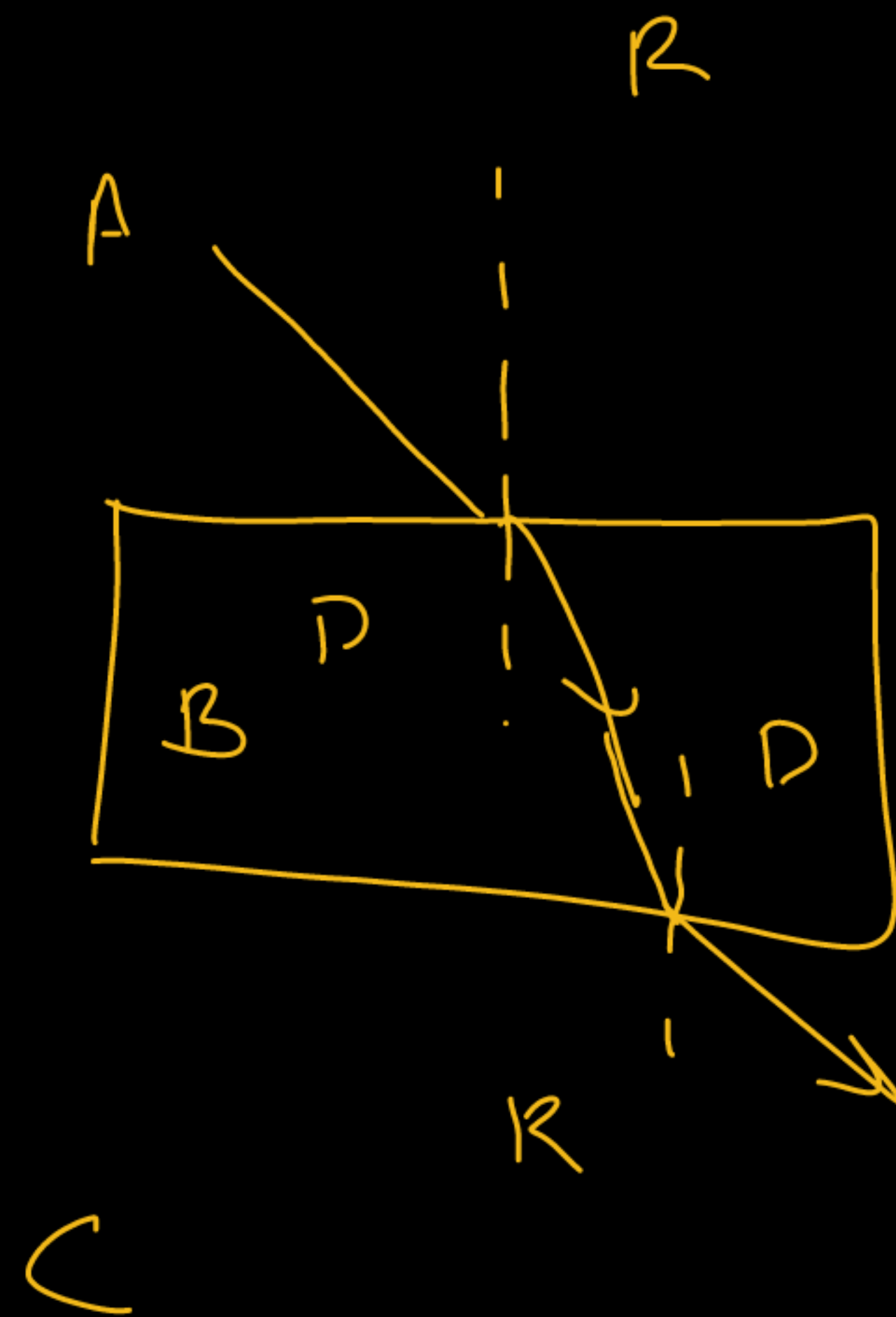
(i) Obliquely and the optical density of medium B is made more than that of A and C.

(ii) The ray falls normally from medium A to medium B. [CBSE 2025]

$$v_c > v_A = v_B$$







Q. The ability of a medium to refract light is expressed in terms of its optical density. Optical density has a definite connotation. It is not the same as mass density. On comparing two media, the one with larger refractive index is optically denser than the other. The other medium with a lower refractive index is optically rarer. Also, the speed of light through a given medium is inversely proportional to its optical density.

(a) Determine the speed of light in diamond if the refractive index of diamond with respect to vacuum is 2.42.

Speed of light in vacuum is 3×10^8 m/s.

(b) Refractive indices of glass, water and carbon disulphide are 1.5, 1.33 and 1.62 respectively.

If a ray of light is incident in these media at the same angle (say i), then write the increasing order of the angle of refraction in these media.

(c) The speed of light in glass is 2×10^8 m/s and in water is 2.25×10^8 m/s.

(i) Which one of the two is optically denser and why?

(ii) A ray of light is incident normally at the water–glass interface when it enters a thick glass container filled with water. What will happen to the path of the ray after entering the glass? Give reason.

HW

Pizza party Question

$$\eta \text{ A WRT B} = \frac{4}{5}$$

$$\eta \text{ B WRT C} = \frac{5}{9}$$

$$\eta \text{ A WRT C} = ?$$

$$\frac{\eta_A}{\eta_C}$$

$$\frac{\eta_A}{\eta_B} = \frac{4}{5}$$

$$\times \frac{\eta_B}{\eta_C} = \frac{5}{9}$$

$$\left(\frac{5}{9} \right)$$

$$\frac{\eta_A}{\eta_B} \times \frac{\eta_B}{\eta_C}$$

- Q. (a) Water has refractive index 1.33 and alcohol has refractive index 1.36. Which of the two medium is optically denser? Give reason for your answer.
- (b) Draw a ray diagram to show the path of a ray of light passing obliquely from water to alcohol.
- (c) State the relationship between angle of incidence and angle of refraction in the above case.
- (d) In which medium will light travel faster? Explain with reason.
- (e) The refractive index of glass is 1.5. The speed of light in vacuum is $3 \times 10^8 \text{ m/s}$. What is the speed of light in glass?

HW